Organism – includes animals, plants and micro organisms.

Population – is a group of similar plants or animals living in an area.

Community – refers to all the plants and animals living in an area.

Ecosystem – all living and non living things and their interaction within an area.

7.3 Ecosystem

Life cannot exist in isolation. It flourishes in an environment which supplies and fulfills its material and energy requirements. A biotic community and its physical environment in which matter and energy flow and cycle is called as ecosystem.

The term *ecosystem* was first proposed by Arthur George Tansley in 1935. Tansley defined ecosystem as, 'the system resulting from the integration of all living and non-living factors of the environment'. The ecosystems can vary in size. It can be very small, extending to about a few square centimeters or it can extend over many square kilometers. Example; tropical forests.

7.3.1 Major components of an ecosystem

The ecosystem is made up of two main components:

A. Abiotic Component and

B. Biotic Component

A. Abiotic Component: This component of the ecosystem includes the non-living substance of the environment. Example; light, air, soil, water, climate, minerals, etc. Sun is the main source of energy for the earth.

B. Biotic Component: This includes a variety of living organisms such as microorganisms, plants and animals. The biotic component of an ecosystem can be further divided into producers, consumers and decomposers based on their capacity to sustain themselves (Figure 7.2).

Ecosystem

https://youtube/aYmdrJWLQ4Y

a. Producers: Organisms that can produce or manufacture their own food are known as producers. Plants that have green pigments or chlorophyll, produce their own food in the presence of CO_2 in the atmosphere, water from the soil and sunlight through a process called '*photosynthesis*'. These green plants are called as '*autotrophs*' (auto – self; trophs – nourishing) as they manufacture their own food.

b. Consumers: Consumers are organisms that cannot manufacture their own food and get their food and nutrients from producers directly or from other organisms. They are called as *'heterotrophs'* (hetero – others; trophs – nourshing).

Consumers can be divided into primary, secondary and tertiary consumers.

1. Primary Consumers

Organisms that feed on producers (green plants) are called primary consumers. They are also called as '*herbivores*' or plant eating organisms. Examples of terrestrial herbivore are

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